Java Assignment: 1

1. What is Java?

Ans: Java is a **high-level, object-oriented programming language** used to build many types of applications—like mobile apps, web apps, desktop software, and even large enterprise systems. Java is a cross platform Object oriented programming language.

2. Is Java platform dependent or independent?

Ans: Java is Platform Independent.

3. What is JVM?

Ans: JVM stands for **Java Virtual Machine**.

How it works

When you write a Java program:

1)Your code is compiled into **bytecode** (not machine code)

2) The JVM reads and executes this bytecode

3) It converts it into machine code that your computer understands

4)What is JDK and JRE?

Ans: JDK (Java Development Kit): A complete package used to develop Java programs. It includes JRE + development tools (compiler, debugger).

JRE (Java Runtime Environment): Used to run Java programs. It includes JVM + libraries.

5) Why is Java called “Write Once, Run Anywhere”?

Ans: Because Java code is compiled into bytecode, which can run on any system that has a JVM (Java Virtual Machine). No need to rewrite code for different platforms.

6) What is a class in Java?

Ans: A class is a blueprint or template used to create objects.

Example -
```
class Student {
    int id;
    String name;
}
```

7) What is an object?

Ans: An object is an instance of a class.

```
Student s1 = new Student();
```

8) What is the main method in Java?

Ans: It is the entry point of a Java program. Execution starts from here.

public static void main(String[] args);

9) What is the use of System.out.println()?

Ans: It is used to print output on the console with a new line.

System.out.println("Hello");

11) What are data types in Java?

Ans: Data types define the type of data a variable can store (like numbers, text, etc.).

12) . Give examples of primitive data types.

Ans: int, double,char,Boolean,byte,short,long

13) . What is a variable?

Ans: A variable is a container used to store data values.

14) How do you declare a variable in Java?

Ans: datatype variableName;

15) What is the difference between int and float?

Ans: int -Stores whole numbers eg.10

Float - Stores decimal numbers eg. 10.5

16) What is type casting?

Ans: Converting one data type into another.

int x = (int) 10.5; // explicit casting

17) What is the default value of int?

Ans: Default value of int is **0** (for instance variables).

18) What is a keyword in Java?

Ans: A keyword is a reserved word with a special meaning (e.g., int, class, if, while).

19) Can we use reserved keywords as variable names?

Ans: No we cannot use keywords as variable names.

20) What is the difference between = and ==?

Ans: Assignment operator(=) Assigns a value.

Comparison operator(==)Compare values.

21) What is an if statement?

Ans: An **if statement** is used to make decisions in a program. It executes a block of code only if a condition is true.

```
if (x > 10) {
 System.out.println("Greater");
 }
```

22) Difference between if and switch?

Ans: IF - Used for multiple conditions — SWITCH - Used for fixed values

Works with all data types — Works with int, char, String, enum

More flexible — More readable for many cases

23) What is a loop?

Ans: A **loop** is used to repeat a block of code multiple times until a condition is satisfied.

24) Types of loops in Java?

Ans: For Loop , While Loop , DO-While Loop

25) Difference between for loop and while loop?

Ans: For loop – Used when number of iterations is known

Initialization, condition, update in one line.

While loop - Used when iterations are unknown

Condition is checked separately.

26) . What is break statement?

Ans : The break statement is used to exit a loop or switch immediately.

27) What is continue statement?

Ans : The continue statement skips the current iteration and moves to the next iteration of the loop.

28) What will happen if we don't write the main method?

Ans : The program will compile successfully, but it will not run.

Error: *Main method not found.*

29) Can we run a Java program without class?

Ans : No. Every Java program must be inside a class.

30) What happens if we don’t compile a Java program?

Ans: You cannot run the program, because Java needs bytecode (.class file) generated after compilation.

31) Can a variable change its data type?

Ans: No A variable’s data type is fixed once declared. (But you can use type casting to convert values.)

32) What is the size of int in Java?

Ans: The size of int is 4 bytes .

33) Why Java does not support pointers directly?

Ans: Java does not support pointers (likes in C/C++)primarily for security ,simplicity and automatic memory management.